

Mansi Aher

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PROFESSIONAL SUMMARY

M.S. Computer Science graduate focused on applied AI and Python backend development. Builds retrieval systems, REST APIs, and data pipelines with FastAPI, PostgreSQL, and Docker; brings research experience in simulation-based machine learning.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, Kotlin, C++, C

Backend & data: FastAPI, SQLAlchemy, REST APIs, PostgreSQL, MySQL, SQLite, pgvector, ChromaDB

Machine learning: PyTorch, scikit-learn, XGBoost, pandas, NumPy, embeddings, RAG, Grad-CAM

Tools: Docker, Docker Compose, Git, GitHub Actions, pytest, Streamlit, GCP, EPANET / WNTR

SELECTED PROJECTS

DevPilot | Semantic Code Search

- Built a full-stack application that searches public GitHub repositories with natural-language questions and returns source snippets with exact file and line references.
- Implemented background indexing, code chunking, and embedding search with FastAPI, PostgreSQL, and pgvector; added authentication, commit-specific citations, and a Docker-based local setup.

Water Leak Detection & AquaTrace | Thesis + Interactive Extension

- Engineered 22 spatial pressure residual features for leak detection on 21,120 simulated LeakDB samples; compared Random Forest, SVM, and XGBoost classifiers.
- Developed AquaTrace from the thesis network: 57 precomputed EPANET scenarios, browser-based Random Forest inference, pressure charts, and reproducible experiment replay; verified inference against 100 Python outputs.

AI Knowledge Copilot | Document Q&A

- Built a retrieval-augmented application that ingests PDFs, generates embeddings, and returns answers with page-level source citations to support verification.
- Separated API routes, services, and persistence; added a pluggable LLM provider interface, Docker Compose services, and automated tests with GitHub Actions.

Brain Tumor MRI Classifier | Computer Vision Prototype

- Implemented a binary tumor / no-tumor classifier using a pretrained ResNet18 model and PyTorch, with MRI preprocessing and a Streamlit inference interface.
- Added Grad-CAM heatmaps to visualize model attention. Developed as an educational prototype without clinical validation.

EXPERIENCE

Research Assistant | California State University, Los Angeles

August 2025 - May 2026

- Developed Python pressure-data processing and model experiments for simulation-based leak detection; analyzed classification metrics and collaborated with faculty on thesis findings.

Web Development Intern | The Sparks Foundation

February 2022 - March 2022

- Designed MySQL schemas and configured IAM access controls on Google Cloud Platform using least-privilege principles.

EDUCATION

M.S., Computer Science | California State University, Los Angeles

August 2024 - May 2026 | GPA: 3.9 / 4.0

B.E., Computer Engineering | Savitribai Phule Pune University

August 2020 - May 2024 | GPA: 3.8 / 4.0